Muscle Mechanics for Aided Diagnosis

Santiago Restrepo Ruiz
Published by Waajacu Open Source Foundation
Santiago.Restrepo.Ruiz@gmail.com https://medical.waajacu.com/

Regional biomechanics and guided movement observation

***Abstract*—We developed Muscles, a browser application that connects interactive anatomy, regional static muscle estimates and structured movement observations. It adapts MS-Human-700 to a client-side MuJoCo runtime, recomputes muscle paths for each posture, and displays activation estimates only when explicit mechanical checks pass. Versioned observation protocols and local reports support reproducible exploration. The system is intended for research and education; its generic model outputs are not patient measurements or diagnoses.**

***Index Terms*—Biomechanics, musculoskeletal modeling, static optimization.**

I. Browser Architecture and Model Integration

The application builds on MS-Human-700 [1] and MuJoCo [2]. Our contribution is the browser integration, regional calculation workflow, interactive visualization and observation/reporting layer. Pinned model files are compiled into browser assets with provenance and integrity hashes. Two documented bilateral path-coordinate corrections are maintained in the imported model.

MuJoCo WebAssembly runs in a Web Worker, while Three.js renders the skeleton and muscle paths. The server delivers static files; model computation runs locally. Users select either arm, either leg, either foot and ankle, trunk, head and neck, or the detailed right hand. Each selection supplies its own coordinates, muscles and posture presets. The primary model supplies eight regions; a separate manipulation profile supplies the articulated right hand. The selected region is emphasized within a complete skeleton for context.

II. Static Muscle Estimation

For each posture, the system resets the model, applies the selected joint coordinates and their equality dependents, and updates tendon paths with their via points, wrapping contacts and pulley segments. Other coordinates remain prescribed. At zero velocity and acceleration, reduced-coordinate force balance includes gravity, passive forces and muscle actuation.

Let a denote muscle activation, $A(q)$ the activation-to-torque map at posture q, $b(q)$ the zero-activation torque baseline, and r unbounded reserve torque. The implemented bounded optimization is

$$\begin{aligned} \min_{a,r} \quad & \|a\|_2^2 + \lambda \|r\|_2^2 \\ \text{subject to} \quad & A(q)a + r = -b(q), \qquad 0 \le a \le 1. \end{aligned}$$

The solver uses $\lambda = 2500\,(\mathrm{N\,m})^{-2}$ and replays the solution through MuJoCo. Activation colors require a maximum absolute residual no greater than $10^{-4}\,\mathrm{N\,m}$, reserve torque no greater than $0.05\,\mathrm{N\,m}$, and passing path, finite-value and capacity checks. Failed checks suppress activation estimates. Reserve use indicates unmet modeled torque demand; it does not establish physiological weakness.

III. Guided Observation and Reports

Guided Assessment selects a body region and, where available, its explicit side, then loads the matching versioned protocol. The right-arm protocol contains 20 distinct postures selected from a deterministic 384-candidate atlas; the final five are supplemental. Selection sought posture variety and distinct generic activation patterns among mechanically acceptable candidates. Eight additional regional protocols extend the observation workflow.

Users record movement completion and symptoms separately from model demand. Local, versioned records support saved reports and matched comparisons; protocol changes archive incompatible observations and exclude them from the active protocol. The main export removes direct identifiers and free text. Six interface languages share stable protocol identifiers and report contracts.

IV. Verification and Scope

Repository verification checks model hashes, regional inventories, protocol identity, localization consistency, deployment files and HTTP behavior. The report suite passes 20 tests, and protocol checks cover nine definitions. These checks establish implementation consistency, not clinical validity.

The calculation describes an unloaded static hold with prescribed support. It does not estimate gait, ground reaction, grip force, fatigue, injury or patient-specific tissue loading. Generic activation is not measured effort, and recorded symptoms do not identify a painful muscle. The application is research and educational software, not a medical device.

Explore the application and its available tool routes at https://medical.waajacu.com/.

References

[1] C. Zuo, K. He, J. Shao, and Y. Sui, "Self model for embodied intelligence: Modeling full-body human musculoskeletal system and locomotion control with hierarchical low-dimensional representation," arXiv:2312.05473v5, 2024. [Online]. Available: https://arxiv.org/abs/2312.05473v5

[2] Google DeepMind, "MuJoCo documentation: Overview," Online documentation, accessed: Sep. 16, 2026. [Online]. Available: https://mujoco.readthedocs.io/en/stable/overview.html